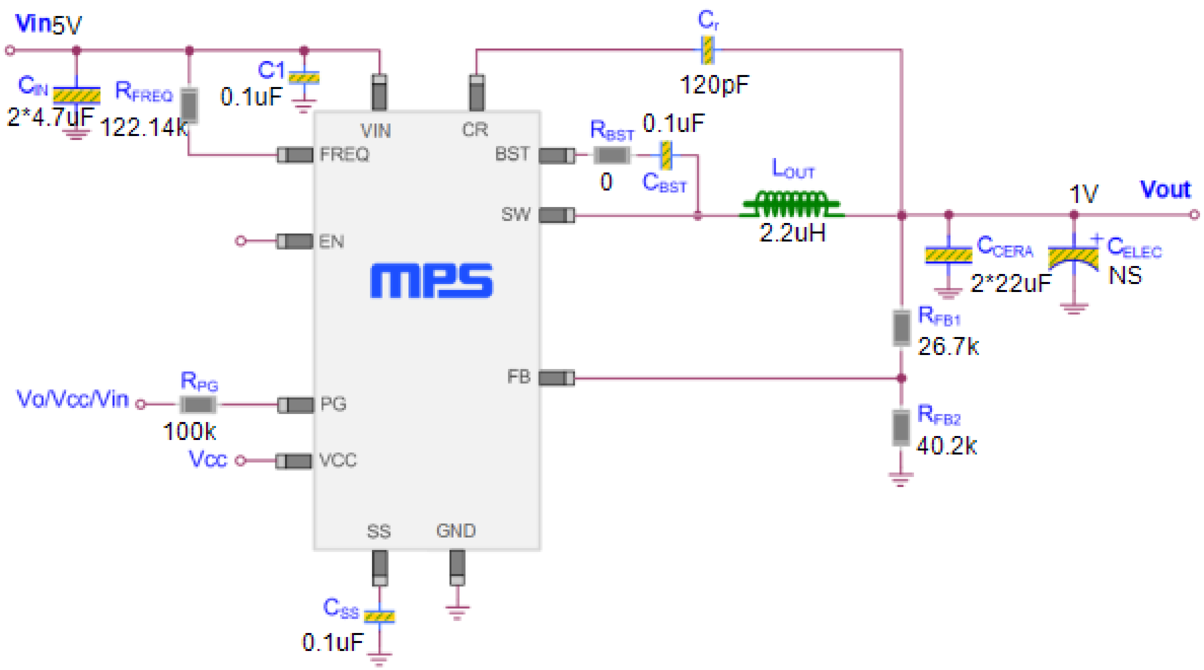
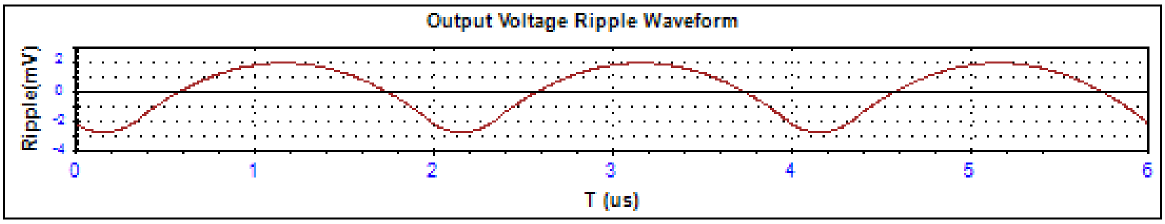
Schematic

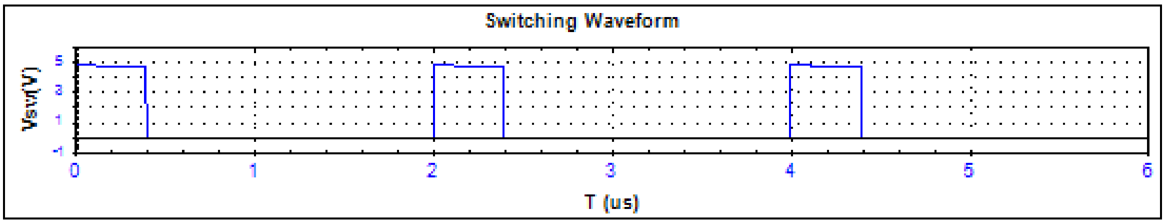

Vin5V
CIN
2*4.7uF
RFREQ
122.14k
C1
0.1uF
VIN
CR
Cr
120pF
FREQ
BST
RBST
0
0.1uF
CBST
SW
LOUT
2.2uH
1V
Vout
EN
CCERA
2*22uF
CELEC
NS
RFB1
26.7k
FB
Vo/Vcc/Vin
RPG
100k
PG
Vcc
VCC
RFB2
40.2k
SS
GND
CSS
0.1uF

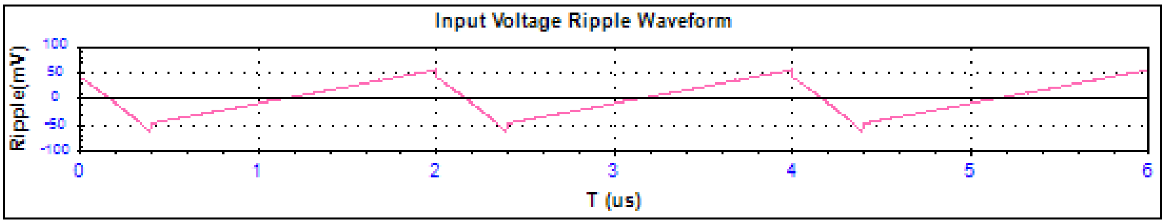
Spec Analysis

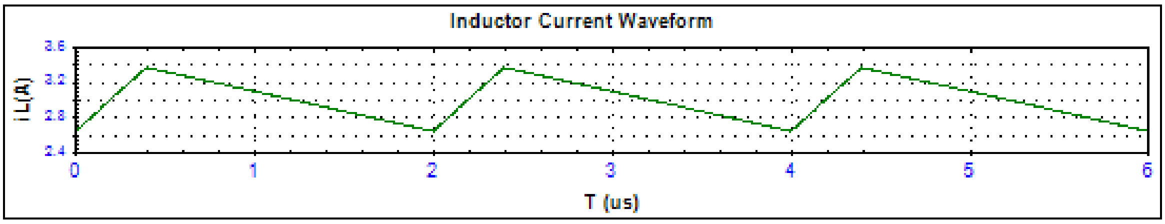
Parameter	Result	Minimum	Maximum
Constant-On-Time Control(Y/N)	Y		
Input Voltage(V)	5		
Min. Input Voltage(V)	5		
Max. Input Voltage(V)	7		
Output Voltage(V)	1		
Output Current(A)	3		
Frequency(kHz)	500		
Slope Ripple (mV)	15		
Input Voltage Ripple(mV)	117.82		
Output Voltage Ripple(mV)	4.64		
Inductor Peak-Peak Current(A)	0.73	2.64	3.36
Efffiency(%)	80.34		

Waveforms

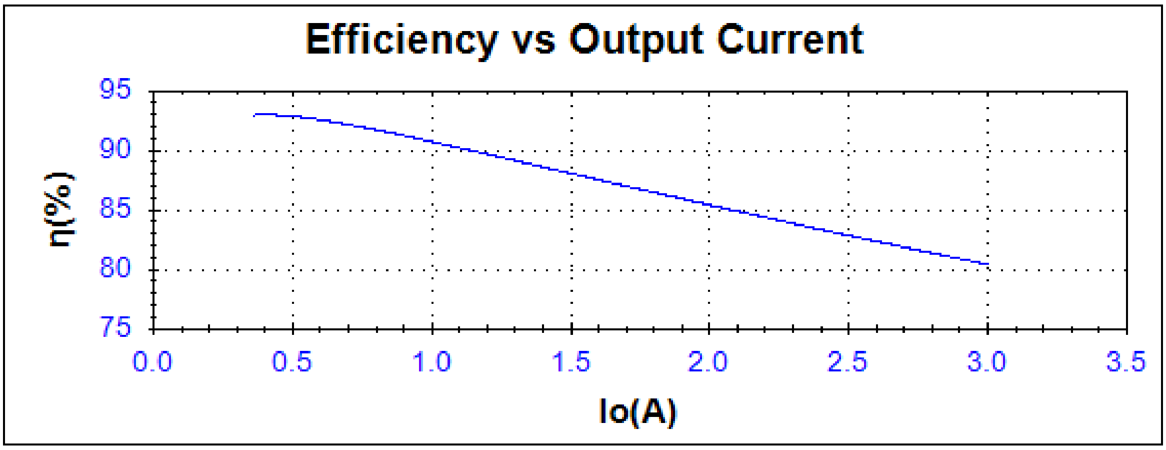

Output Voltage Ripple Waveform
Ripple(mV)
2
0
-2
-4
0
1
2
3
4
5
6
T (us)

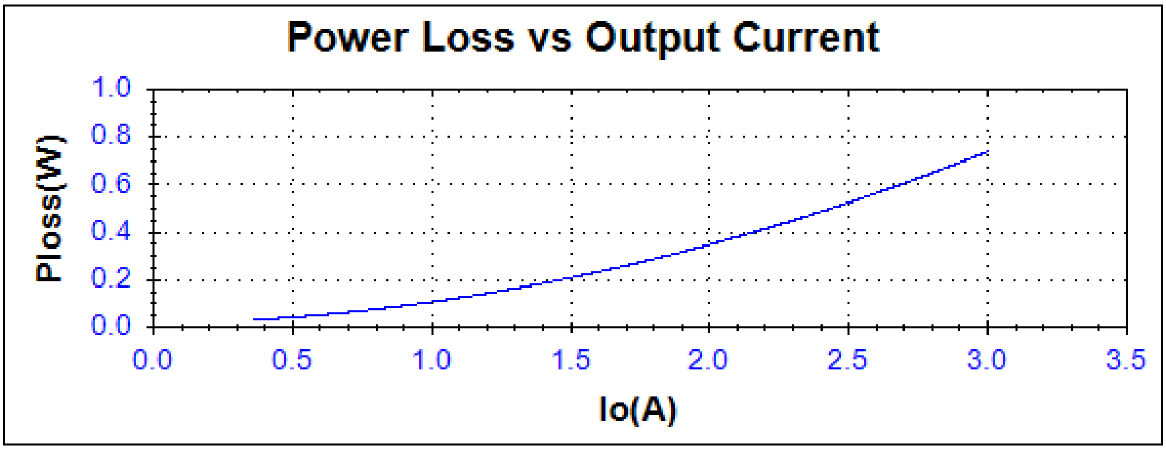

Switching Waveform
Vsw(V)
5
3
1
-1
0
1
2
3
4
5
6
T (us)


Input Voltage Ripple Waveform
Ripple(mV)
100
50
0
-50
-100
0
1
2
3
4
5
6
T (us)


Inductor Current Waveform
iL(A)
3.6
3.2
2.8
2.4
0
1
2
3
4
5
6
T (us)

Efficiency & Power Loss Analysis





Bill of Materials

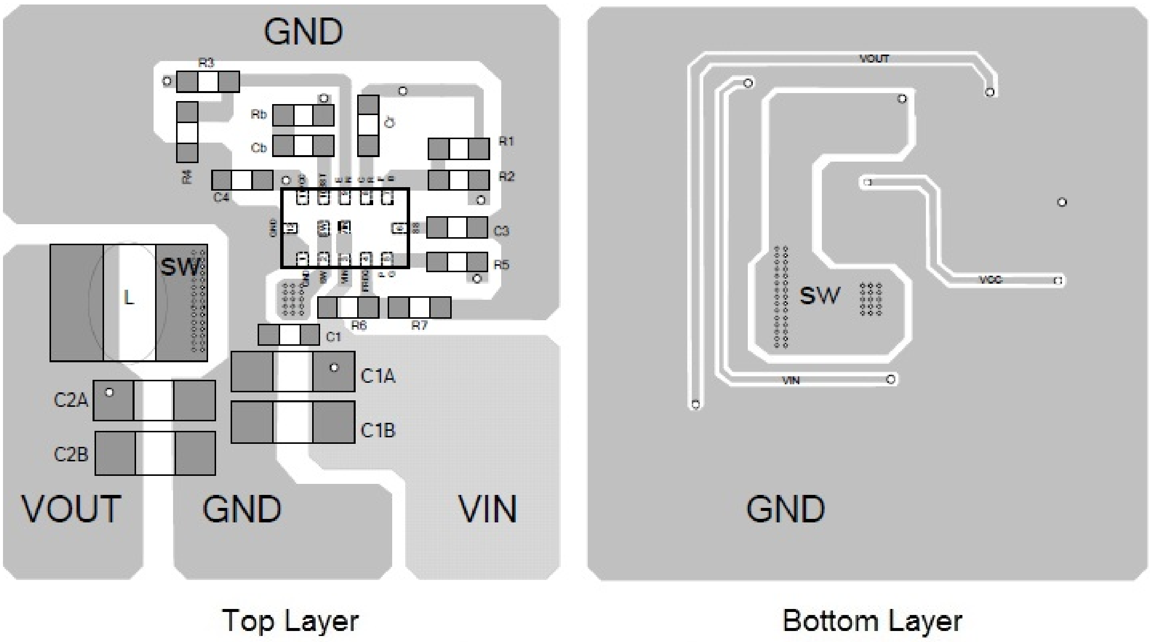
Quantity	Parameter	Value	Description	Package	Vendor	Part Number
2	Cin	4.7uF	10V, X5R, Ceramic	0402	Wurth Elektronik	885012105026
1	C1	0.1uF	50V, Ceramic	0603	Any	
2	Ccera	22uF	6.3V, X5S, Ceramic	0805	muRata	GRM21BC60J226M
	Celec		NS			
1	Cr	120pF	50V, , Ceramic	0603	Any	
1	CSS	0.1uF	50V, Ceramic	0603	Any	
1	CBST	0.1uF	50V, Ceramic	0603	Any	
1	RPG	100k	5%	0603	Any	
1	RBST	0	5%	0603	Any	
1	RFB1	26.7k	1%	0603	Any	
1	RFB2	40.2k	1%	0603	Any	
1	RFREQ	122.14k	1%	0603	Any	
1	LOUT	2.2uH	32mOhm,6.7A	SMT	Wurth Elektronik	7443733448022
1	U1	-	Cot, 500kHZ, 19V, 4A	QFN2x3-14	MPS	MP2326

Note:
1) More details information please reference the device datasheets and/or application notes.
2) Please note too large output capacitance may lead to startup problems.

Layout Info

PCB layout is very important to achieve stable operation. Please follow these guidelines and take the figures for reference. The layout figure is from datasheet, please refer to datasheet 's typical application circuit as corresponding schematic.

1) Keep the power loop of input capacitor, HS switch, LS switch as small as possible.
2) Keep the connection of input capacitor and IN pin as short and wide as possible.
3) Ensure all feedback connections are short and direct. Place the feedback resistors and compensation components as close to the chip as possible.
4) Route SW away from sensitive analog areas such as FB.
5) Connect IN, SW, and especially GND respectively to a large copper area to cool the chip to improve thermal performance and long-term reliability.
6) Adding RC snubber circuit from IN pin to SW pin can reduce SW spikes.



MP2326 PCB Layout